

Yixiang Gao

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Education

- 2017 – Current 📖 **Ph.D. Electrical and Computer Engineering**, University of Missouri - Columbia.
Thesis: Confounded predictions in machine learning - *Detect and mitigate confounding factors in machine learning and deep learning models*.
(Expected Graduation: July 2024)
- 2014 – 2017 📖 **B.S. Computer Engineering**, University of Missouri - Columbia.
B.S. Electrical Engineering, University of Missouri - Columbia.

Skills

- Research 📖 Deep Learning, Computer Vision, Object Detection, Robotics, Speech Processing, sEMG
Pattern Recognition, Machine Learning on Embedded Devices
- Coding 📖 Python, MATLAB, CUDA, C++, C
- Libraries 📖 pytorch, scikit-learn, ONNX, mmdet, ultralytics, ROS, tinygrad
- Languages 📖 English, Chinese.

Experience

- 2017 – Current 📖 **Graduate Research Assistant**, Vision-Guided and Intelligent Robotics Laboratory
- Neurobiological and Psychobiological Signatures of Vocal Effort in Early Career Teachers
Built machine learning tools to analysis voice acoustic signals, sEMG, and fMRI to understand neurological and biological causes for muscle tension voice disorders.
 - Neck Surface EMG Signals for the Early Detection of Vocal Fatigue in Student Teachers
Built the data collection protocol in both hardware and software for this study. Used classifiers such as SVM, MLP, and CNN to detect vocal fatigue from collected data.
 - Object Detection and Pose Estimation using Embedded Devices
Deployed YOLO algorithm on a Raspberry Pi with two stereo cameras for object detection and pose estimation and produced comparable results to state-of-art deep learning pose estimation model.
 - Built Spotmicro - a 3D-printed Robotic Dog
Built the first iteration of SpotMicro for MU Robotics club. Created a platform to help undergraduate and graduate students to learn, practice what they have learned in class.

Research Publications

Journal Articles

- 1 Y. Gao, M. Dietrich, and G. N. DeSouza, "Classification of vocal fatigue using semg: Data imbalance, normalization, and the role of vocal fatigue index scores," *Applied Sciences*, vol. 11, 2021.

Conference Proceedings

- 1 Y. Gao, M. Berardi, M. Dietrich, and G. N. DeSouza, "Removal of confounding factors using ga-svm feature adaptation: Application on detection of vocal fatigue thru semg classification," in *IEEE 2023 Congress on Evolutionary Computation (CEC)* [Accepted], 2023.

- 2 J. Demby's, **Y. Gao**, and G. N. DeSouza, "A study on solving the inverse kinematics of serial robots using artificial neural network and fuzzy neural network," in *2019 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*, 2019.
- 3 J. Demby's, **Y. Gao**, A. Shafiekhani, and G. N. DeSouza, "Object detection and pose estimation using cnn in embedded hardware for assistive technology," in *2019 IEEE Symposium Series on Computational Intelligence (SSCI)*, 2019.
- 4 **Y. Gao**, M. Dietrich, M. Pfeiffer, and G. N. DeSouza, "Classification of semg signals for the detection of vocal fatigue based on vfi scores," in *2018 40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2018.

Conference Presentations

- 1 **Y. Gao**, M. Berardi, M. Dietrich, and G. N. DeSouza, "Feature adaptation with ga-svm for confounding removal - an application on vocal fatigue detection using semg classification," The 15th Advances in Quantitative Laryngology, Voice and Speech Research (AQL), 2023.
- 2 **Y. Gao**, M. Dietrich, and G. N. DeSouza, "Classification of vocal fatigue using neck semg with leave-one-subject-out testing," The 14th Advances in Quantitative Laryngology, Voice and Speech Research (AQL), 2021.
- 3 **Y. Gao**, M. Dietrich, and G. N. DeSouza, "Explore voice production variability through neck semg clustering - challenge for accurate labeling of vocal fatigue," The 14th Advances in Quantitative Laryngology, Voice and Speech Research (AQL), 2021.
- 4 M. Dietrich, **Y. Gao**, and G. N. DeSouza, "Extralaryngeal surface emg features that distinguish between those with and without elevated scores on the vocal fatigue index," The Fall Voice Conference, 2019.
- 5 M. Dietrich, E. Tippit, A. Walker, M. Pfeiffer, **Y. Gao**, and G. N. DeSouza, "Relative fundamental frequency during vocal loading and relationships with laryngeal muscular patterns," The 13th Pan-European Voice Conference, 2019.
- 6 **Y. Gao**, M. Dietrich, M. Pfeiffer, A. Walker, and G. N. DeSouza, "Classification of vocal gestures extracted from quasi-daily sentences to detect vocal fatigue," The 13th Advances in Quantitative Laryngology, Voice and Speech Research (AQL), 2019.
- 7 **Y. Gao**, M. Pfeiffer, M. Dietrich, and G. N. DeSouza, "Classification of neck surface emg signals for the early detection of vocal dysfunction," The 12th Advances in Quantitative Laryngology, Voice and Speech Research (AQL), 2017.